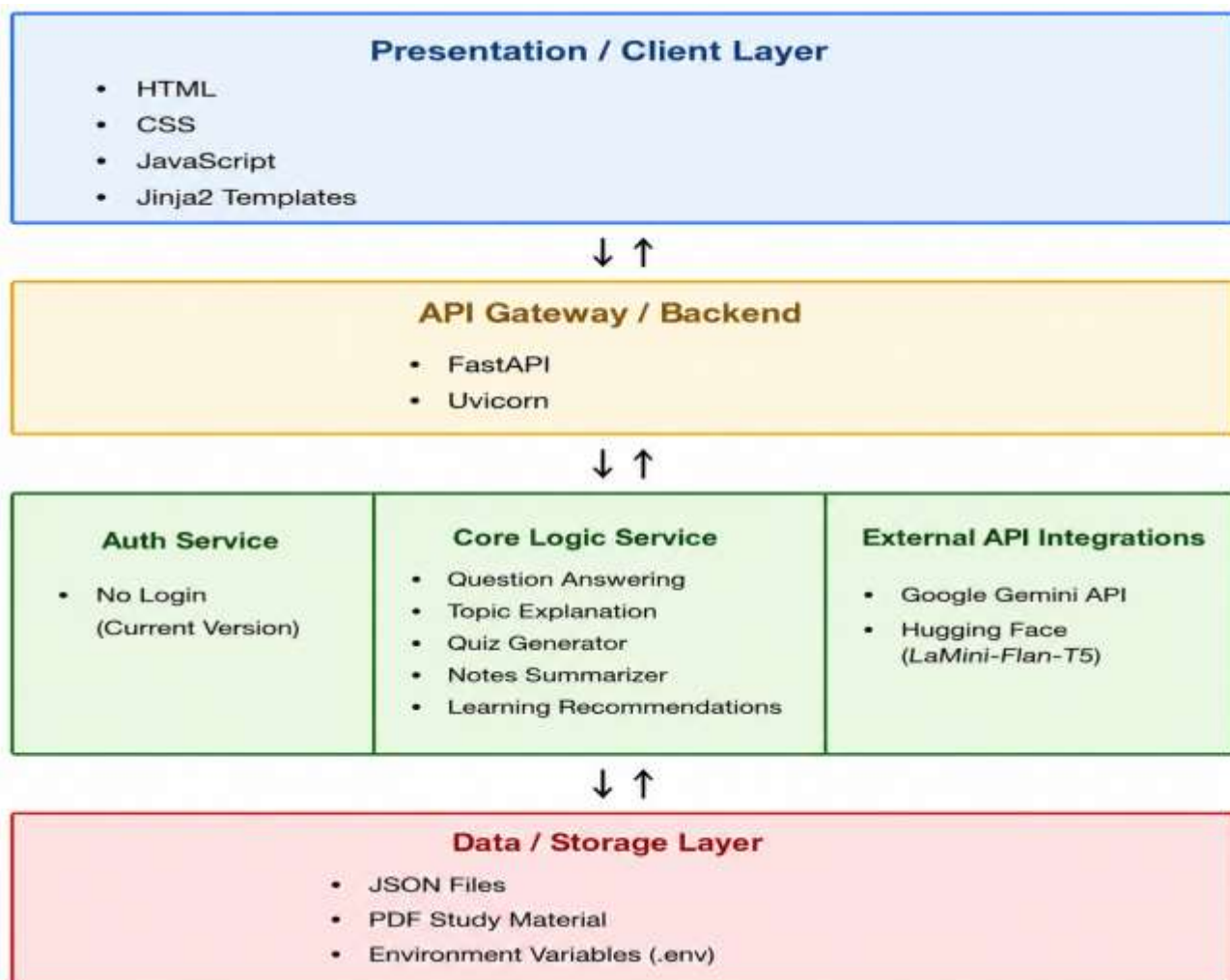


Solution Architecture

Date	13 July 2026
Team ID	XXXXXX
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the web interface where students can ask questions, generate quizzes, view summaries, explanations, and recommendations.</i>	HTML, CSS, JavaScript, Jinja2 Templates
<i>API Gateway/ Backend</i>	<i>Receives HTTP requests from the frontend, validates input, routes requests to the correct AI module, and returns responses.</i>	FastAPI, Uvicorn
AI Processing Services	Executes Question Answering, Topic Explanation, Quiz Generation, Notes Summarization, and Learning Recommendation modules using AI models.	Python, Google Gemini API, Hugging Face (LaMini-Flan-T5)
External AI Services	<i>Connects with external AI providers to generate intelligent responses and educational content.</i>	Google Gemini API, Hugging Face Transformers
Data / Storage Layer	Stores uploaded PDFs, configuration files, application settings, and generated data for processing.	JSON Files, PDF Study Material, .env Environment Variables
<i>Testing Layer</i>	<i>Verifies backend APIs, AI modules, and application functionality to ensure reliable performance.</i>	PyTest, FastAPI TestClient